

2.3 集合的积与无交并

定理(无交并的“外在”版本的泛性质刻画) $(A_i)_{i \in I}$ 是任意一族集合.

A 是一个集合, $(\iota_i: A_i \longrightarrow A)_{i \in I}$ 是一族映射, 满足:

* 每个 ι_i 都是单射,

* $A = \bigcup_{i \in I} \text{im}(\iota_i)$

* 对 $\forall i, j \in I, i \neq j$, 有: $\text{im}(\iota_i) \cap \text{im}(\iota_j) = \emptyset$.

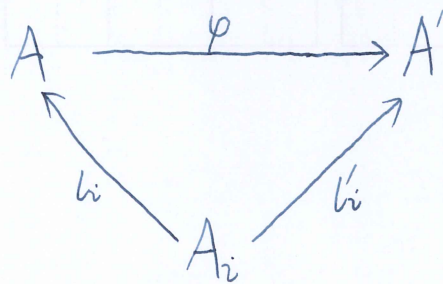
A' 是另一个集合, $(\iota'_i: A_i \longrightarrow A')_{i \in I}$ 是另一族映射, 满足:

* 每个 ι'_i 都是单射,

* $A' = \bigcup_{i \in I} \text{im}(\iota'_i)$

* 对 $\forall i, j \in I, i \neq j$, 有: $\text{im}(\iota'_i) \cap \text{im}(\iota'_j) = \emptyset$.

则有: 存在唯一的映射 $\varphi: A \longrightarrow A'$, 使得对 $\forall i \in I$, 下图交换:



且映射 φ 是双射.

Proof: 对 $\forall x \in A$. 我们按照下面的方式来定义映射 $\varphi: A \rightarrow A'$.

$$\because x \in A = \bigcup_{i \in I} \text{im}(l_i) \quad \therefore \exists t \in I, \text{ s.t. } x \in \text{im}(l_t)$$

$\therefore l_t$ 是 $A_t \rightarrow A$ 的映射

$$\therefore \exists \alpha \in A_t, \text{ s.t. } l_t(\alpha) = x$$

$\therefore l'_t$ 是 $A_t \rightarrow A'$ 的映射

$$\therefore l'_t(\alpha) \in A'$$

定义 $\varphi(x) := l'_t(\alpha) \in A'$. 显然有 $\varphi(A) \subseteq A'$.

对 $\forall x_1, x_2 \in A$. 有:

$$\because x_1 \in A = \bigcup_{i \in I} \text{im}(l_i) \quad \therefore \exists t_1 \in I, \text{ s.t. } x_1 \in \text{im}(l_{t_1})$$

$\therefore l_{t_1}$ 是 $A_{t_1} \rightarrow A$ 的映射

$$\therefore \exists \alpha_1 \in A_{t_1}, \text{ s.t. } l_{t_1}(\alpha_1) = x_1$$

$\therefore l'_{t_1}$ 是 $A_{t_1} \rightarrow A'$ 的映射

$$\therefore l'_{t_1}(\alpha_1) \in A'$$

$$\therefore \varphi(x_1) = l'_{t_1}(\alpha_1) \in A'$$

$$\because x_2 \in A = \bigcup_{i \in I} \text{im}(l_i) \quad \therefore \exists t_2 \in I, \text{ s.t. } x_2 \in \text{im}(l_{t_2})$$

$\therefore l_{t_2}$ 是 $A_{t_2} \rightarrow A$ 的映射

$$\therefore \exists \alpha_2 \in A_{t_2}, \text{ s.t. } l_{t_2}(\alpha_2) = x_2$$

$\therefore l'_{t_2}$ 是 $A_{t_2} \rightarrow A'$ 的映射

$$\therefore l'_{t_2}(\alpha_2) \in A'$$

$$\therefore \varphi(x_2) = l'_{t_2}(\alpha_2) \in A'.$$

若 $x_1 = x_2$, 则有: $x_1 \in \text{im}(l_{t_1}), x_1 = x_2 \in \text{im}(l_{t_2})$

$$\therefore x_1 \in \text{im}(l_{t_1}) \cap \text{im}(l_{t_2}) \quad \therefore t_1 = t_2$$

$$\therefore l_{t_1}(\alpha_1) = x_1 = x_2 = l_{t_2}(\alpha_2) = l_{t_1}(\alpha_2) \quad \therefore l_{t_1} \text{ 是单射} \quad \therefore \alpha_1 = \alpha_2$$

$$\therefore \varphi(x_1) = l'_{t_1}(\alpha_1) = l'_{t_2}(\alpha_1) = l'_{t_2}(\alpha_2) = \varphi(x_2) \quad \therefore \varphi \text{ 是 } A \rightarrow A' \text{ 的映射.}$$

若 $\varphi(x_1) = \varphi(x_2)$, 则有: $l'_{t_1}(\alpha_1) = l'_{t_2}(\alpha_2)$

$\therefore l'_{t_1}(\alpha_1) \in \text{im}(l'_{t_1}), \quad l'_{t_1}(\alpha_1) = l'_{t_2}(\alpha_2) \in \text{im}(l'_{t_2})$

$\therefore l'_{t_1}(\alpha_1) \in \text{im}(l'_{t_1}) \cap \text{im}(l'_{t_2}) \quad \therefore t_1 = t_2$

$\therefore l'_{t_1}(\alpha_1) = l'_{t_2}(\alpha_2) = l'_{t_1}(\alpha_2) \quad \therefore l'_{t_1} \text{ 是单射} \quad \therefore \alpha_1 = \alpha_2$

$\therefore x_1 = l_{t_1}(\alpha_1) = l_{t_2}(\alpha_1) = l_{t_2}(\alpha_2) = x_2 \quad \therefore \varphi \text{ 是 } A \rightarrow A' \text{ 的单射.}$

对 $\forall d \in A', \quad \therefore d \in A' = \bigcup_{i \in I} \text{im}(l'_i) \quad \therefore \exists k \in I, \text{ s.t. } d \in \text{im}(l'_k)$

$\therefore l'_k \text{ 是 } A_k \rightarrow A' \text{ 的映射} \quad \therefore \exists \gamma \in A_k, \text{ s.t. } l'_k(\gamma) = d$

$\therefore l_k \text{ 是 } A_k \rightarrow A \text{ 的映射} \quad \therefore l_k(\gamma) \in A$

$\therefore \varphi(l_k(\gamma)) = l'_k(\gamma) = d \quad \therefore \varphi \text{ 是 } A \rightarrow A' \text{ 的满射.}$

$\therefore \varphi \text{ 是 } A \rightarrow A' \text{ 的双射.}$

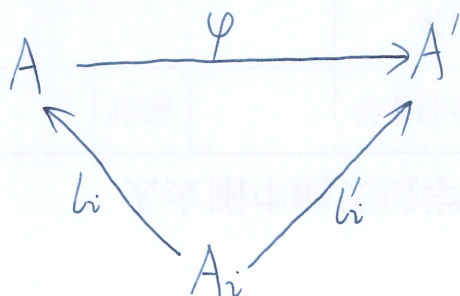
对 $\forall i \in I, \quad \therefore l_i \text{ 是 } A_i \rightarrow A \text{ 的映射, } \varphi \text{ 是 } A \rightarrow A' \text{ 的映射}$

$\therefore \varphi \circ l_i \text{ 是 } A_i \rightarrow A' \text{ 的映射.}$

$\therefore i \in I \quad \therefore l'_i \text{ 是 } A_i \rightarrow A' \text{ 的映射.}$

对 $\forall \varepsilon \in A_i$, 有: $(\varphi \circ l_i)(\varepsilon) = \varphi(l_i(\varepsilon)) = l'_i(\varepsilon)$

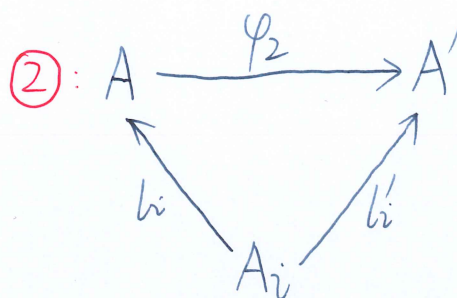
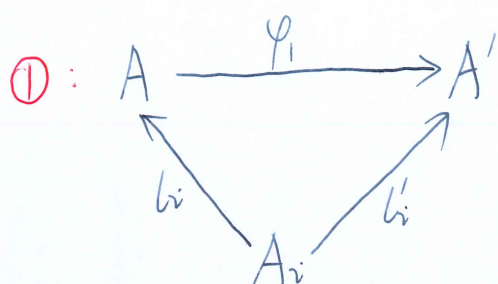
$\therefore \varphi \circ l_i = l'_i \quad \therefore \text{下图交换:}$



$\therefore \text{存在性得证.}$

下证唯一性. 假设存在两个映射 $\varphi_1: A \rightarrow A'$, $\varphi_2: A \rightarrow A'$,

使得: 对 $\forall i \in I$, 下图交换:



对 $\forall \theta \in A$. $\because \theta \in A = \bigcup_{i \in I} \text{im}(l_i) \quad \therefore \exists q \in I, \text{ s.t. } \theta \in \text{im}(l_q)$

$\because l_q$ 是 $A_q \rightarrow A$ 的映射 $\therefore \exists \mu \in A_q, \text{ s.t. } l_q(\mu) = \theta$

$\therefore \varphi_1(\theta) = \varphi_1(l_q(\mu)) = (\varphi_1 \circ l_q)(\mu) = l'_q(\mu)$

图①交换

$= (\varphi_2 \circ l_q)(\mu) = \varphi_2(l_q(\mu)) = \varphi_2(\theta)$

图②交换

$\therefore \varphi_1 = \varphi_2$

$\therefore$ 唯一性得证

